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Chapter 17

Switching Technologies

The ICND2 exam expects you to be comfortable with switching technologies such as VLANs, Spanning Tree Protocol (STP), VLAN Trunking Protocol (VTP), Etherchannel, and port security. Some of these concepts were introduced in the ICND1 exam. The ICND2 exam expects you to know not only the purpose of each of these protocols and technologies, but also how to configure and troubleshoot them! Spend some time practicing the configuration of these technologies, and then answer the following questions.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding Spanning Tree Protocol (STP)
- Configuring VLANs
- Configuring VLAN Trunking Protocol (VTP)
- Implementing Etherchannel
- Managing port security

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Spanning Tree Protocol is a layer 2 protocol designed to prevent layer 2 loops on the network. It does this by deciding which link should be placed in a blocking state. STP takes time to converge as each port on a switch goes through many states, such as blocking, listening, learning, and forwarding.
- Rapid Spanning Tree Protocol (RSTP) converges quickly by selecting alternate and potential backup ports. When a link goes down, RSTP doesn't reconverge; it simply activates one of the alternate or backup ports. This saves a lot of time!
- Know how to configure VLANs and port security on a Cisco switch. Virtual LANs are communication boundaries that allow you to separate your network even on a single switch. Know that trunk links are special ports that can carry traffic for all VLANs.
- VLAN Trunking Protocol (VTP) is a protocol that is used to simplify the management of VLANs across switches in the enterprise. With VTP, you create the list of VLANs on the VTP server and then that list is sent to any VTP clients configured for the same VTP domain. A VTP domain is a group of switches that will share VLAN information.

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Understanding Spanning Tree Protocol (STP)

700–716 Choose the best answer(s).

700. In regards to STP, what is the default priority on a Cisco switch?

(A) 32,768

(B) 16,384

(C) 8,192

(D) 4,096

701. Which layer of the OSI model does RSTP run at?

(A) Physical

(B) Data Link

(C) Network

(D) Transport

702. Which of the following is a benefit of RSTP over STP?

(A) RSTP transitions to a forwarding state faster than STP does.

(B) RSTP has more port states than STP.

(C) RSTP uses the same port roles that STP uses.

(D) RSTP can route between VLANs.

703. Which of the following terms describes a spanning tree network where all switch ports are in either forwarding or blocking state?

(A) Redundant

(B) Converged

(C) Fault tolerant

(D) Treed

704. Which of the following is selected by STP as the root bridge?

(A) The switch with the highest priority

(B) The switch with the highest bridge ID

(C) The switch with the lowest IP address configured

(D) The switch with the lowest bridge ID

705. PVST+ introduced which of the following port states?

(A) Forwarding

(B) Learning

(C) Discarding

(D) Listening

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706. Which of the following represent STP port states? (Select three.)

(A) Blocking

(B) Listening

(C) Disabled

(D) Forwarding

707. Looking at the output of the show spanning-tree command displayed in the figure below, is SW1 the root bridge?

```
SW1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    4097
             Address     00E0.F72C.AEC0
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4097  (priority 4096 sys-id-ext 1)
             Address     00E0.F72C.AEC0
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/24           Desg FWD 19        128.24   P2p
```

(A) Yes

(B) No

708. Looking at the following switch information, how can you configure SwitchC as the root bridge?

Name: SwitchA

Priority: 32768

MAC: 00-00-0c-00-b0-01

Name: SwitchB

Priority: 32768

MAC: 00-50-0d-10-00-00

Name: SwitchC

Priority: 32768

MAC: 0b-3f-27-00-93-3a

(A) Increase the priority.

(B) Lower the priority.

(C) Change the MAC address.

(D) Change the name.

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709. Once the root switch has been selected, each switch must then determine its root port. Which port is selected as the root port?

(A) The port with the lowest MAC address

(B) The port with the highest MAC address

(C) The port with the lowest cost

(D) The port with the highest cost

710. Which of the following are true regarding RSTP? (Select three.)

(A) RSTP uses the same port roles as STP

(B) Reduces converging time after a link failure

(C) STP transitions quicker to a forwarding state than RSTP

(D) Uses additional port roles over STP

(E) Transitions to a forwarding state faster than STP

711. You would like to verify if your switch is the root bridge. What command would you use?

(A) `show stp`

(B) `show spanning-tree`

(C) `show tree`

(D) `show root bridge`

712. You have three switches connected in a loop with the configuration of each switch shown below. Which switch will be selected as the root bridge?

Name: SwitchA
Priority: 32768
MAC: 00-00-0c-00-b0-01

Name: SwitchB
Priority: 32768
MAC: 00-50-0d-10-00-00

Name: SwitchC
Priority: 32768
MAC: 0b-3f-27-00-93-3a

(A) SwitchA

(B) SwitchB

(C) SwitchC

(D) All of them

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713. Which of the follow commands was used to create the output in the figure below?

```
Name: Fa0/1
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Appliance trust: none
```

(A) `show stp`

(B) `show spanning-tree`

(C) `Show interface fastethernet 0/1`

(D) `show interface fastethernet 0/1 switchport`

714. Which of the following identifies additional port roles used by RSTP?

(A) Designated

(B) Root

(C) Blocking

(D) Alternate

715. You would like to ensure that when a system connects to a switch that it receives connectivity right away with little wait time. Which of the following commands would you use?

(A) (config-if)#`spanning-tree portfast`

(B) (config-if)#`spanning-tree lowwait`

(C) (config)#`spanning-tree portfast`

(D) (config-if)# `portfast`

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716. Using the figure below, what do you need to do on switch SW2 in order to ensure it is the root bridge?

```
SW2#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     00E0.F72C.AEC0
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000D.BD54.01EE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/24           Altn BLK 19        128.24   P2p
```

(A) Add an additional interface.

(B) Increase the aging time.

(C) Increase the priority above 4096.

(D) Decrease the priority below 4096.

Configuring VLANs

717–726 Choose the best answer(s).

717. Which of the following statements are true of VLANs and their usage? (Select two.)

(A) Communication between VLAN requires a router.

(B) You cannot use VLANs across switches.

(C) A VLAN that spans across switches requires a router.

(D) Each VLAN requires its own IP subnet.

(E) Multiple VLANs can use the same IP subnet if sub-interfaces are used on the router.

718. Which of the following are VLAN tagging protocols?

(A) ISL

(B) 802.1d

(C) 802.1q

(D) 802.11

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719. You would like to create a VLAN called MKT on your switch. Which command(s) would you use?

(A) Use the following commands:

```
vlan 10
name MKT
```

(B) Use the following command:

```
vlan MKT
```

(C) Use the following command:

```
vlan name MKT
```

(D) Use the following command:

```
vlan name MKT num 10
```

720. Which of the following statements is true of the native VLAN?

(A) Both sides of the trunk link should use the same native VLAN.

(B) Both sides of the trunk link should use different native VLANs.

(C) Native VLAN is only used when the frame is tagged VLAN 1.

(D) Native VLANs are only used to tag voice traffic.

721. You want to ensure that only VLAN 10 and VLAN 20 traffic can pass over a trunk link. Which command would you use?

(A) SW1(config-if)#`trunk allowed vlan 10,20`

(B) SW1(config)#`switchport trunk allowed vlan 10,20`

(C) SW1(config-if)#`switchport trunk allowed vlan 10,20`

(D) SW1(config)#`trunk allowed vlan 10,20`

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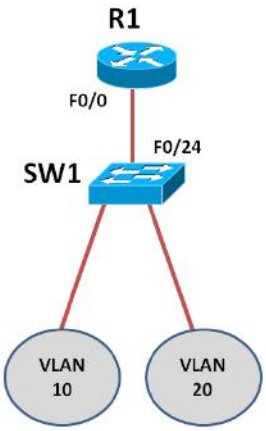
722. Looking at the figure below, which of the following statements are true about the interVLAN routing?



(A) VTP must be enabled on R1

(B) F0/0 must be configured with sub-interfaces

(C) RIP must be enabled on SW1

(D) F0/0 on R1 and F0/24 on the switch must use the same encapsulation protocol

(E) F0/24 on the switch must be configured with sub-interfaces

723. Two switches are connected together by a crossover cable. The ports connecting the two switches have been configured for access mode. You have systems on each switch that are part of VLAN 20 and VLAN 30. What is your assessment of this configuration?

(A) Systems will not be able to communicate between the two switches.

(B) Systems in VLAN 20 can communicate across the switches.

(C) Systems in VLAN 30 can communicate across the switch.

(D) Change the cable from a crossover cable to a straight-through in order to allow systems to communicate between the two switches.

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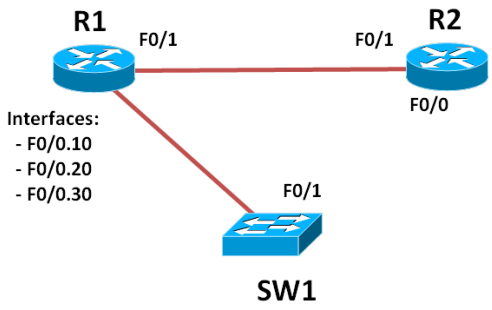
724. Using the figure below, which of the following statements are true of router R1?



(A) There are too many sub-interfaces on f0/0.

(B) Interface f0/0 should be configured as an access port.

(C) Interface f0/0 should be configured as a trunk port.

(D) Interface F0/1 should be configured as a trunk port.

725. You have created VLAN 10 and would like to place Fast Ethernet port 0/8 in VLAN 10. Which command(s) would you use?

(A) Use the following command:

sw(config)#`switchport access vlan 10`

(B) Use the following commands:

sw(config)#`interface f0/8`

sw(config-if)# `access vlan 10`

(C) Use the following commands:

sw(config)#`interface f0/8`

sw(config-if)#`switchport access vlan 10`

(D) Use the following commands:

sw(config)#`interface f0/8`

sw(config-if)#`switchport vlan 10`

726. Which of the following port types cannot be ISL trunk ports?

(A) 1 Gbps

(B) 100 Mbps Fast Ethernet

(C) 10 Mbps Ethernet

(D) 10 Gbps

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Configuring VLAN Trunking Protocol (VTP)

727–734 Choose the best answer(s).

727. Which protocol is used to facilitate the management of VLANs?

(A) STP

(B) 802.1q

(C) ISL

(D) VTP

728. Which VTP mode does not allow the creating of VLANs?

(A) Client mode

(B) Server mode

(C) Transparent mode

(D) Parent mode

729. You wish to prevent the sending of VLAN traffic to other switches if the destination switch does not have any ports in a specific VLAN. What VTP feature would you enable?

(A) VTP domain

(B) VTP pruning

(C) VTP password

(D) VTP passphrase

730. You would like to configure switch SW2 to receive VLAN information for the glensworld VTP domain. Which command(s) would allow you to do this?

(A) Use the following commands:

```
vtp domain P@ssw0rd
vtp password glensworld
vtp mode server
```

(B) Use the following commands:

```
vtp domain glensworld
vtp password P@ssw0rd
vtp mode transparent
```

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(C) Use the following commands:

```
vtp domain glensworld
vtp password P@ssw0rd
vtp mode client
```

(D) Use the following command:

```
vtp mode client
```

731. Which VTP mode allows the creation of VLANs but does not accept changes from other VTP systems and does forward VTP messages on to other devices?

(A) Server mode

(B) Client mode

(C) Transparent mode

(D) Parent mode

732. Which of the following commands configures a port on the Cisco switch for trunking using 802.1q protocol?

(A) Use the following commands:

```
sw(config-if)#switchport mode trunk
sw(config-if)#encapsulation dot1q
```

(B) Use the following commands:

```
sw(config-if)#switchport mode trunk
sw(config-if)#encapsulation isl
```

(C) Use the following commands:

```
sw(config-if)#switchport mode trunk
sw(config-if)#switchport trunk encapsulation isl
```

(D) Use the following commands:

```
sw(config-if)#switchport mode trunk
sw(config-if)#switchport trunk encapsulation dot1q
```

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733. You have configured SW1 as a VTP server for the GlensWorld VTP domain and a password of P@ssw0rd. You use the following commands to configure SW2 as a VTP client, but are unsuccessful. What is the problem?

```
vtp domain glensworld
vtp password P@ssw0rd
vtp mode client
```

(A) The password needs changing.

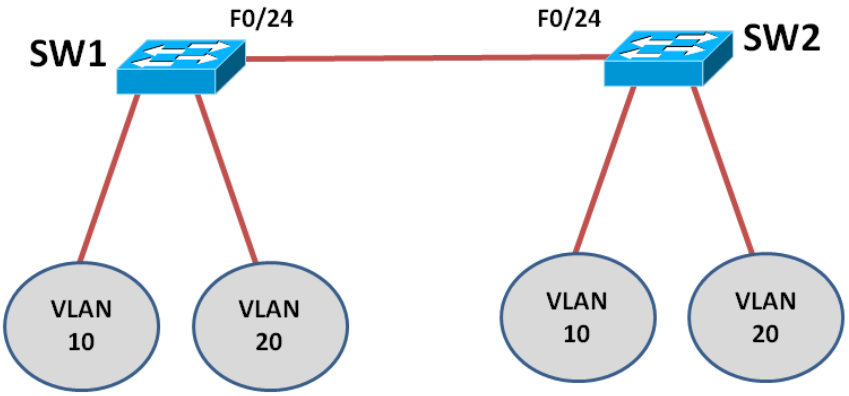
(B) The VTP mode should be set to transparent.

(C) The VTP domain should be unique.

(D) The VTP domain name is case sensitive.

734. You have typed the following command on switch SW1. Using the figure below, what effect will the commands have on the network?

```
Interface f0/24
Switchport mode access
```



(A) All systems in VLAN 10 can communicate with one another.

(B) Systems will not be able to communicate between the two switches.

(C) All systems in VLAN 20 can communicate with one another.

(D) Systems can communicate between the two switches.

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Implementing Etherchannel and Port Security

735–737 Choose the best answer(s).

735. You wish to configure an Etherchannel link made up of two Fast Ethernet trunk ports without any negotiations. What command would you use when configuring the interfaces?

(A) `channel-group 1 mode active`

(B) `channel-group 1 mode passive`

(C) `channel-group 1 mode off`

(D) `channel-group 1 mode on`

736. Which of the following is true in regards to the following commands?

```
Interface fa0/8
Switchport mode access
Switchport port-security
Switchport port-security maximum 3
Switchport port-security mac-address 1111.2222.3333
```

(A) The switch will learn three MAC addresses and associate them with port 8.

(B) The switch will learn eight MAC addresses and associate them with port 3.

(C) The system with the MAC address of 1111.2222.3333 can connect to port 8 along with three other systems.

(D) The system with the MAC address of 1111.2222.3333 can connect to port 8 along with two other systems.

737. You have configured three trunk ports in an Etherchannel group. What will happen when one port in the grouping fails?

(A) The channel cost is increased.

(B) The channel cost is decreased.

(C) STP places the grouping in a blocking state.

(D) STP places the grouping in a forwarding state.